
4. Troubleshooting

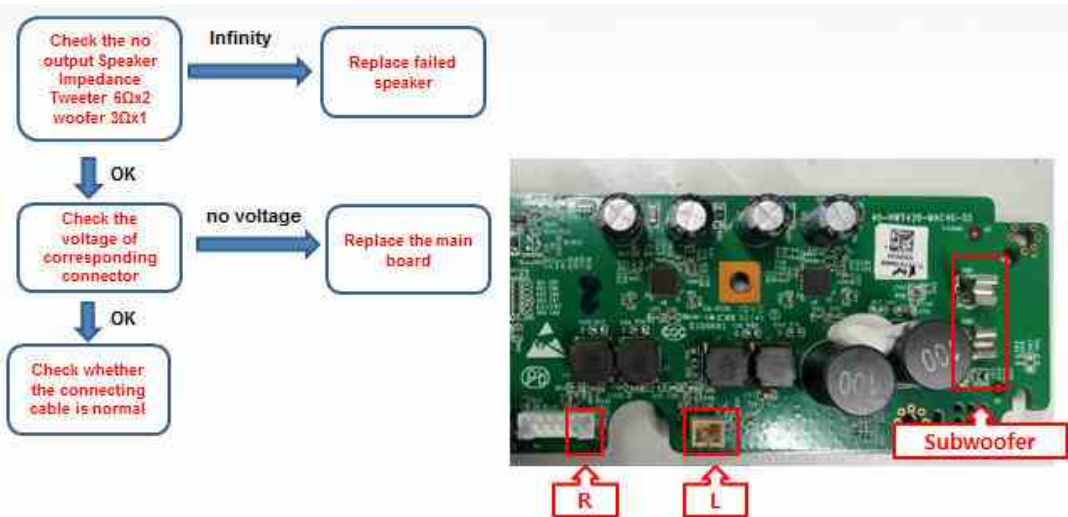
4-1. Checkpoints by Error Mode

Oscilloscope Setting Values	Check DC voltage	Check I ² C& I ² S signal	check PWM signal	Check L+,L- signal
Voltage/DIV	Depend voltage	1V/div	10V/div	10V/div
TIME/DIV	500ms/div	20us/div	1us/div	500us/div

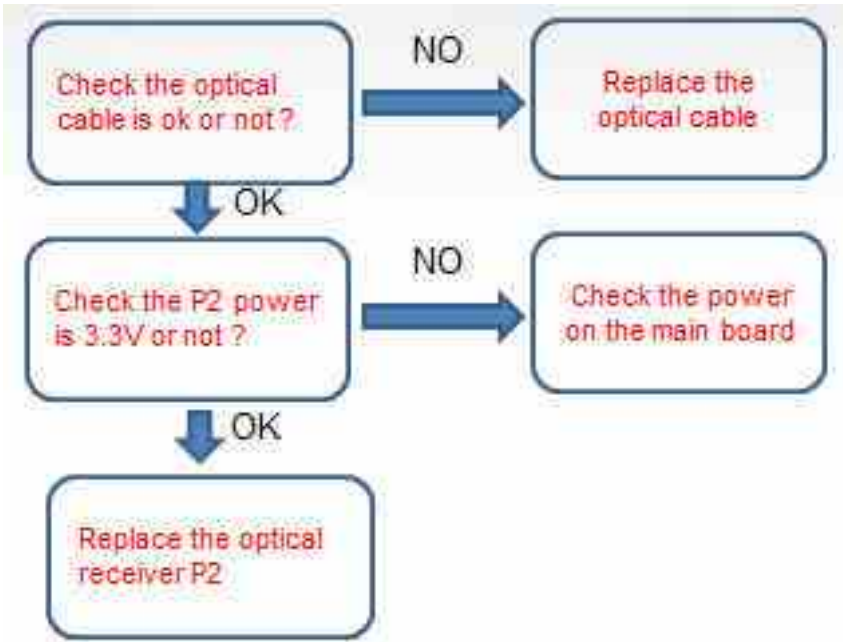
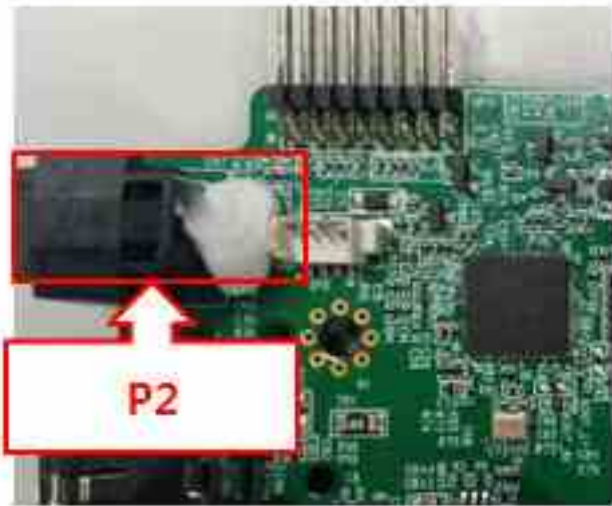
4-1-1. No Power

Symptom	- If you connect the power cables, the LED on the front will turn on - If the units do not work
Major checkpoints	In case of no power
Diagnostics	<pre> graph TD A[Check pin 22 to 24 voltage of XP1 on the main board] -- "< 22V" --> B[Check pin 1 /2to 3/4 voltage of XP6 on the power board] A -- "22-24V" --> C[Check the output of CVB4 on the main board] B -- "no voltage" --> D[Check is the power cord connected properly] B -- "AC voltage" --> E[Replace the power board] C -- "< 4.5V" --> F[Replace the UB1 on main board] C -- "4.5-5.5V" --> G[Check the output of CBK1 on the main board] G -- "< 3.1V" --> H[Replace the UBK1 on main board] G -- "3.1-3.5V" --> I[U6 on the main board fault Replace the main board] I -- "NO" --> J[Replace the VFD Board] </pre> The diagnostic flowchart for 'No Power' includes the following steps and actions: Step 1: Check pin 22 to 24 voltage of XP1 on the main board. - If voltage is $< 22V$, proceed to Step 2. - If voltage is 22-24V, proceed to Step 3. Step 2: Check pin 1 /2to 3/4 voltage of XP6 on the power board. - If no voltage, check if the power cord is connected properly. - If AC voltage, replace the power board. Step 3: Check the output of CVB4 on the main board. - If voltage is $< 4.5V$, replace the UB1 on main board. - If voltage is 4.5-5.5V, proceed to Step 4. Step 4: Check the output of CBK1 on the main board. - If voltage is $< 3.1V$, replace the UBK1 on main board. - If voltage is 3.1-3.5V, U6 on the main board fault. Replace the main board. - If NO, replace the VFD Board. Two images are provided for reference: - An image of a power cord connector labeled XP6. - An image of the main board with components labeled: U6, CBK1, CBV4, UBK1, UB1, and XP1.
Caution	Before working, disconnect the power cord.

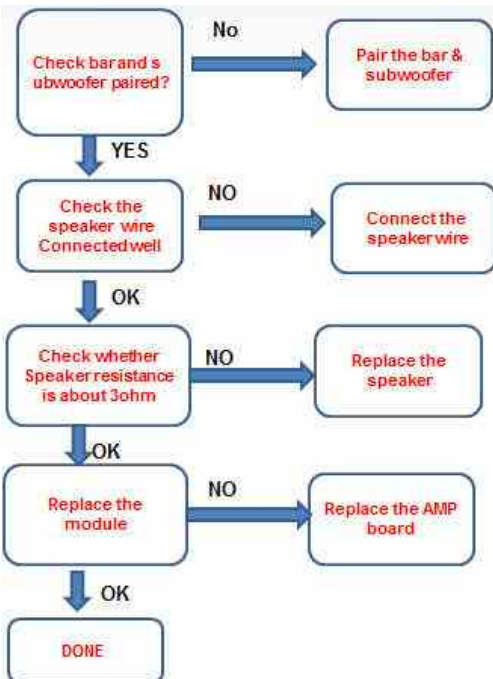
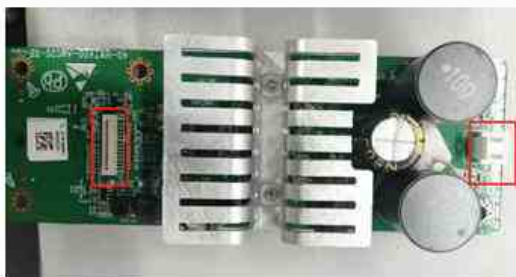
4-1-2. No Audio Output

Symptom	In case of no sound
Major checkpoints	- Speaker connector is not connected properly or check for damage. - Make sure your sound processing functions operate properly on the motherboard. - Make sure the speaker isn't bad.
Diagnostics	 The diagnostics section contains a flowchart and a photograph of a motherboard. Flowchart: <pre> graph TD A[Check the no output Speaker Impedance Tweeter 6Ωx2 woofer 3Ωx1] -- Infinity --> B[Replace failed speaker] A -- OK --> C[Check the voltage of corresponding connector] C -- no voltage --> D[Replace the main board] C -- OK --> E[Check whether the connecting cable is normal] </pre> Image: A photograph of a green motherboard. Red boxes highlight specific components: two small square components labeled 'R' and 'L' (likely speaker connectors), and a larger component labeled 'Subwoofer'.
Caution	Before working, disconnect the power cord.

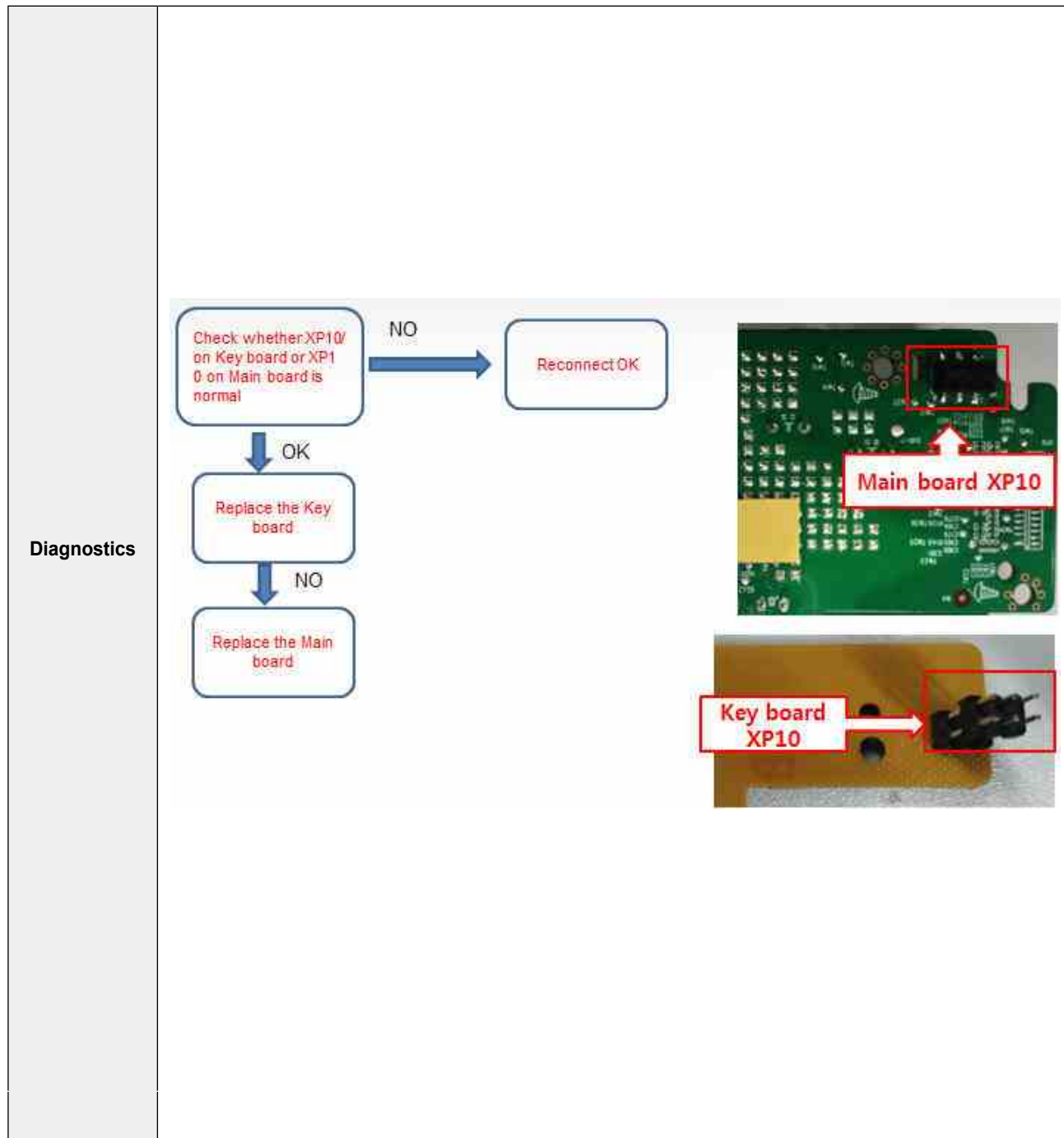
4-1-3. Optical is not working

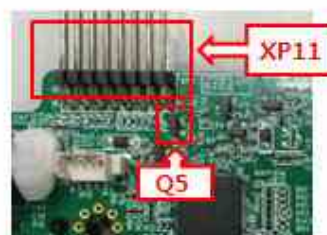
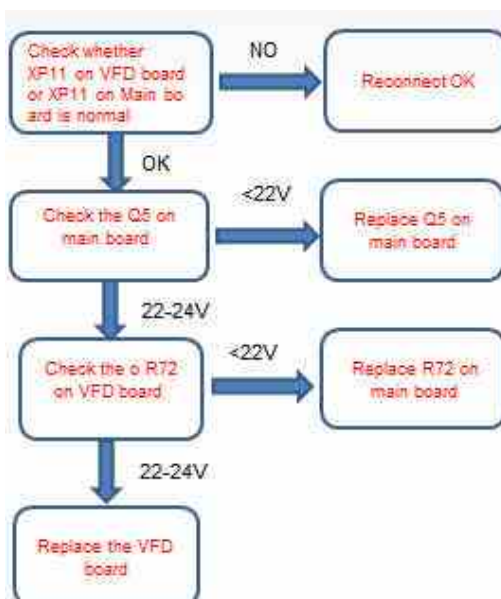
Symptom	In case of optical is not working
Major checkpoints	
Diagnostics	 <pre> graph TD A[Check the optical cable is ok or not?] -- NO --> B[Replace the optical cable] A -- OK --> C[Check the P2 power is 3.3V or not?] C -- NO --> D[Check the power on the main board] C -- OK --> E[Replace the optical receiver P2] </pre> 
Caution	Before working, disconnect the power cord.

4-1-4. No sound output(Subwoofer)

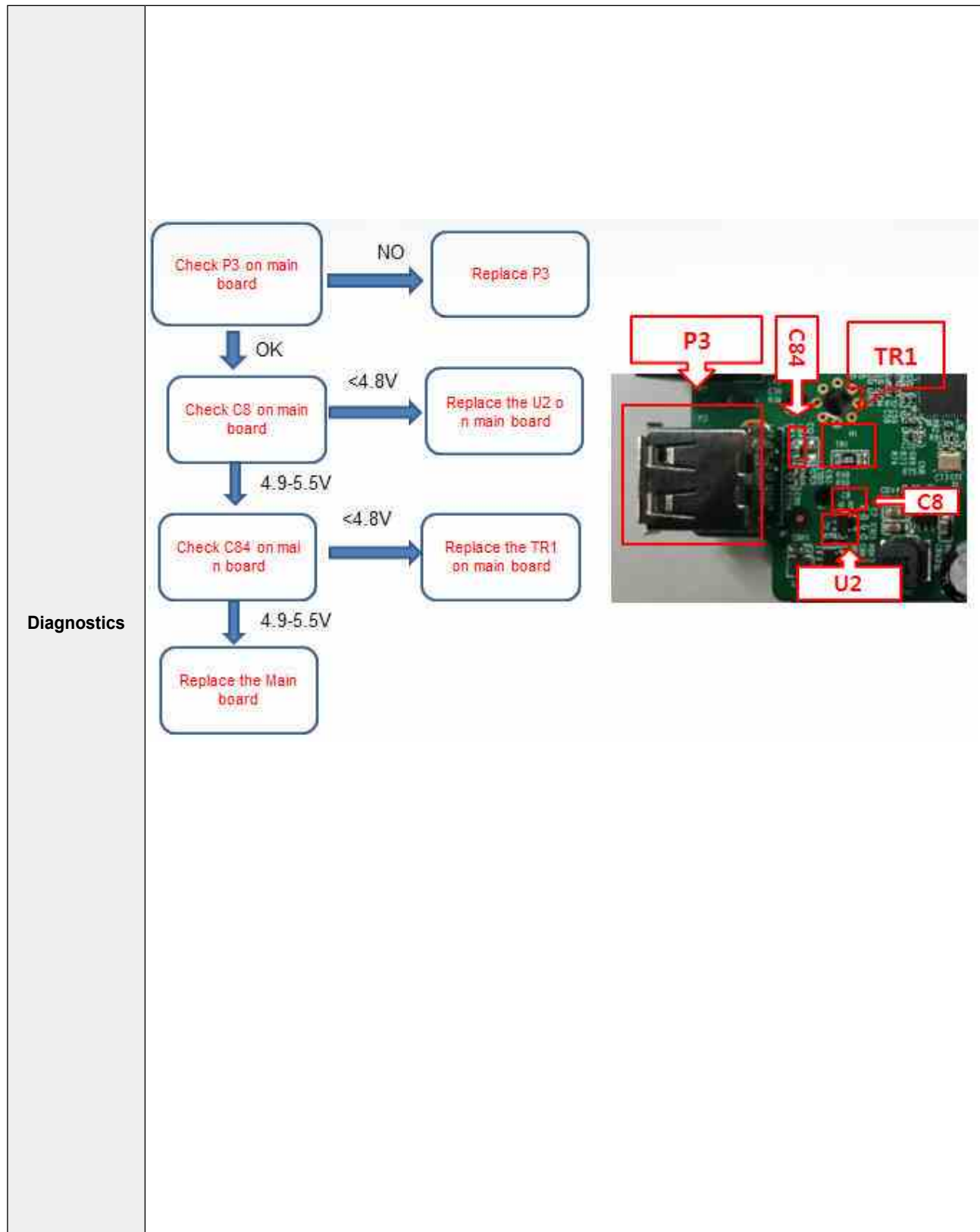
Symptom	In case of no sound(subwoofer)
Major checkpoints	
Diagnostics	 <pre> graph TD A[Check bar and subwoofer paired?] -- No --> B[Pair the bar & subwoofer] A -- YES --> C[Check the speaker wire Connected well] C -- NO --> D[Connect the speaker wire] C -- OK --> E[Check whether Speaker resistance is about 3ohm] E -- NO --> F[Replace the speaker] E -- OK --> G[Replace the module] G -- NO --> H[Replace the AMP board] G -- OK --> I[DONE] </pre> 
Caution	Before working, disconnect the power cord.

4-1-5. In case of key is not working



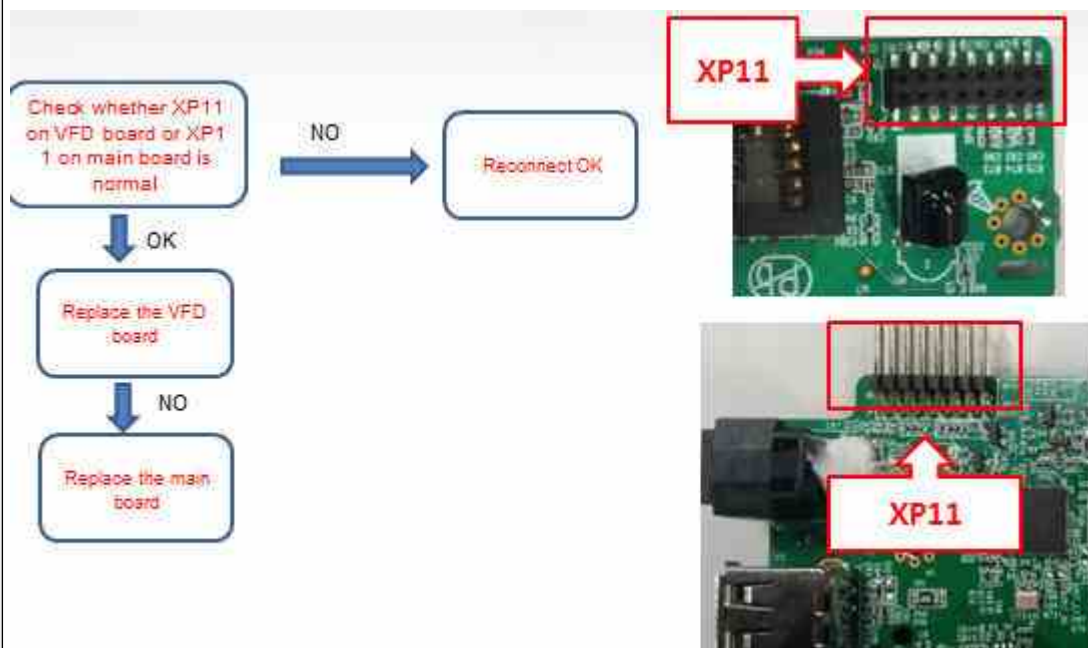
4-1-6. In case of VFD is not working**Diagnostics**

4-1-6. In case of USB is not working



4-1-7. In case Bluetooth is not working

Diagnostics



4-2. Measures to be taken when the Protection Circuit operates

4-2-1. AMP Pre-Inspection relating to Power Protection

If you think there are problems at the AMP PCB, you can check the PCB without disassembling the set by following the test below.

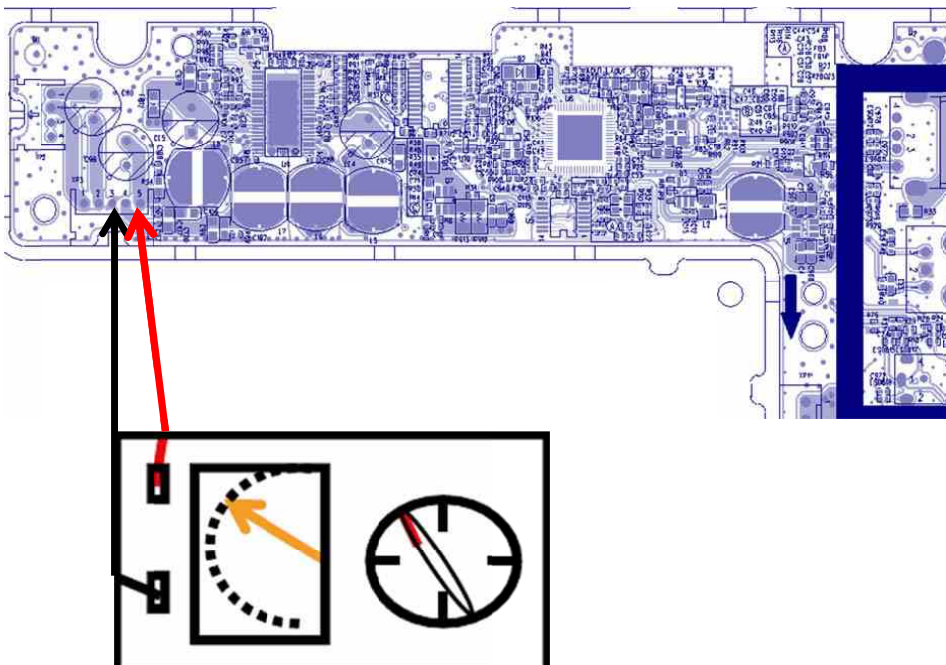


Caution

- Do not connect the power cord during the test!

Resistance using Tester	
F/R CH	9.48M ohm

- If there is a large difference than the value listed above then the AMP PCB has a problem.



4-3. Initialization & Update

4-3-1. Set Initialization method

Initialization

Service or Factory Initial: clear BT link and use history

1. Turn the unit on
2. Press and hold the “+” and “-” button on Sound bar more than 5s
3. “INIT” appear on VFD , about 3s.
4. If initialization successful, “OK” appear on VFD and then unit turn off.

Customer Initial: no clear BT link and use history

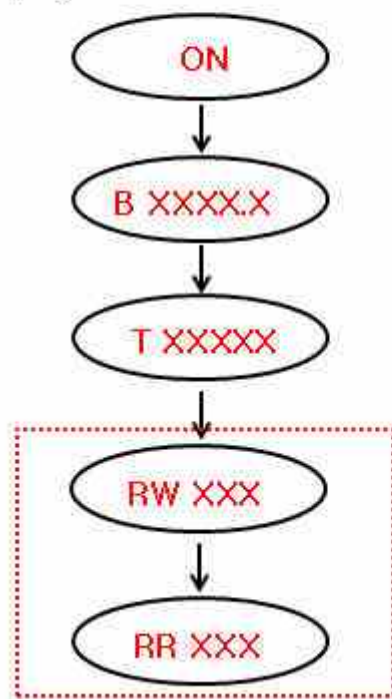
1. Turn the unit off
2. Press and hold the “PLAY/PAUSE” button on Remote or Sound bar more than 5s
3. If initialization successful, “INIT” appear on VFD , about 3s.

4-3. Initialization & Update

4-3-2. How to check firmware version

. Software check method

- Sound Bar(using the remote)
- 1) Prepare the remote control.
- 2) Turn the unit off
- 3) Press the "  " button on the remote for 5 seconds.
- 4) OLED will display each versions like below sequence, (Backend→Tx→RW→RR(If connected))



Tx : Wireless Tx module in the soundbar
 RW : Wireless Rx module in the Subwoofer kit
 RR : Wireless Rx module in the Rear Speaker kit
 (if connected the kit)

B : Backend version(Representative Ver.)
 T : Wireless Tx Version
 RW : Wireless SubWoofer Version
 RR : Wireless Rear Speaker Version
 (if connected the kit)

Subwoofer VERSION DISPLAY
 - No Sub : RW ---
 - Sub Connected : RW S33

Rear VERSION DISPLAY
 - No Rear : RR ---
 - Rear Connected : RR S33

4-3-3. USB Update procedure

COPY Update File:

- ◆ Copy update file to USB root, Make sure no others file in USB root.
- ◆ File need to be named as below table shows.

Module	Updated SW file name
HW-T450	HMS-T4X0WWB-xxxx.ROM

Step1:

1. Clear USB flash driver, Copy "HMS-T4X0WWB-xxxx.ROM" into USB flash driver root.
2. Plug USB flash driver into USB terminal of bar, power on T450 sound bar.
3. Switch source to USB, then VFD will display "UPDATE" and an progress form 0~100.
4. After SW update is finished, Bar will restart automatically.
5. [Check version](#) (click and jump to chapter check version) the bar version shall be "B xxxxx".

Step2(Upgrade Wireless Rx):

Note: you must upgrade Wireless Rx first then upgrade wireless Tx.

1. Connect wireless woofer, confirm the link led of woofer is light on (blue constant).
2. Clear you disc, Copy T450 Rx module software(HMS-WWRRX-XXXX.BIN) you selected in U disc root, plug U disc into bar, and switch source to USB.
3. Then VFD will display "UPDATE" and a progress form Rx 0~ Rx 100.
4. After SW update is finished, Bar will shut down.

Step3(Upgrade Wireless Tx):

1. Connect wireless woofer, confirm the link led of woofer is light on (blue constant).
2. Clear you disc, Copy T450 Tx module software you selected (HMS-WWRTX-XXXX.BIN) in U disc root, plug U disc into bar, and switch source to USB.
3. Then VFD will display "UPDATE" and an progress form Tx 0~ Tx 100.
4. After SW update is finished, Bar will shut down.

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